Fleet condition report: La Haute Borne wind farm

La Haute Borne wind farm (4x Senvion MM82 2050 kW, France), ENGIE open data via NREL OpenOA. 4 turbines, SCADA 01 Jan 2014 to 31 Dec 2015; assessment window: the last 24 months of each turbine (Jan 2014 to Dec 2015). Generated by `windml report` from the mapping `la_haute_borne.yaml`.

Executive summary

All 4 turbines are green: no validated warning sign in the data up to 31 Dec 2015. Green is not a guarantee; see the notes and limits below.

Green: LHB01, LHB02, LHB03, LHB04.

How the ratings work. A turbine is amber only for a specific, unusual pattern: gear-oil pump pressure building up far beyond its sister turbines, repeated oil-system or pitch trips, or a faulty sensor. It is red when the pattern is strong (e.g. a pump pressure rise of 2 bar or more) or persistent and still active. Thresholds were chosen on 2017-2020 data from two UK wind farms and checked on 2021-2024: amber or red turbine-months were followed by a logged fault, trip or long service stop within 90 days 76% of the time, against 50% for random turbines in the same months; but only 19% of the turbine-months before such an event carried a flag. Ratings help decide where to look first; most faults still arrive without a warning sign in SCADA data. This export has no alarm log, so the trip rules could not run.

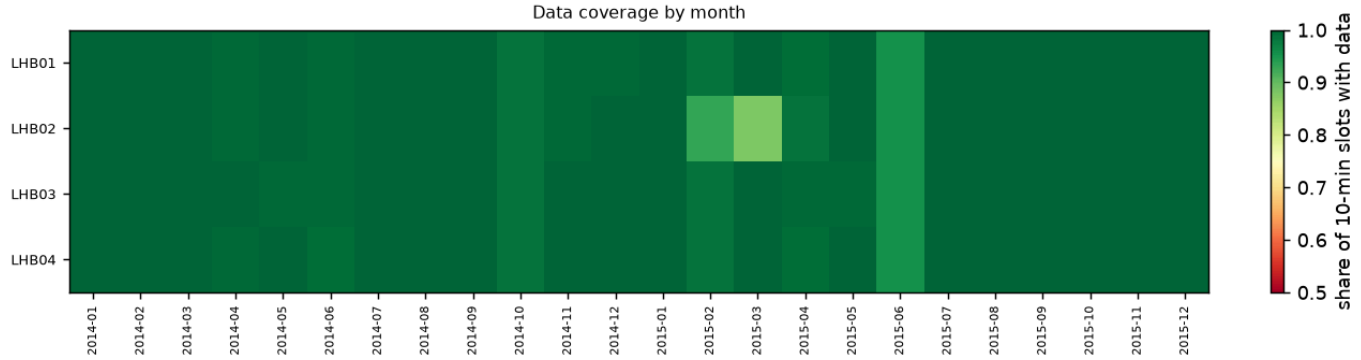
How sure are we. Confidence is *High* when the turbine has 3+ years of history, near-complete data, models trained on earlier years, an alarm log and at least 3 sister turbines to compare with; it drops when any of these is missing. A red rating is a reason to look, not a diagnosis.

What this report could not check:

- There is **no alarm log**, so we cannot see faults, trips or availability. Stops are inferred from SCADA (no power while the wind blows), and the trip rules cannot run.
- With neither an alarm log nor gear-oil pump pressure, only the sensor checks can change a rating for this fleet: green here mostly means 'no sensor problem found'. Model alerts are still shown as notes.
- The export has no gearbox oil temperatures, no oil pressures, no main-bearing temperatures, no pitch-motor temperatures, so those components could not be checked.
- Models scored only about 50% of the window (the first year of a record is needed to train them).

1. Data quality

Turbine	First data	Last data	Months	Coverage (window)	Longest gap (h)	Signals	Native interval	Frozen rows (window)	Stuck sensors (>= 1%)	Stop/fault labels
LHB01	2014-01-01	2015-12-31	24	99.5%	34	3	10min	0	-	inferred from SCADA (no alarm log)
LHB02	2014-01-01	2015-12-31	24	98.8%	133	3	10min	146	-	inferred from SCADA (no alarm log)
LHB03	2014-01-01	2015-12-31	24	99.6%	34	3	10min	42	-	inferred from SCADA (no alarm log)
LHB04	2014-01-01	2015-12-31	24	99.6%	34	3	10min	0	-	inferred from SCADA (no alarm log)



Signals present in at least one turbine: `ambient_temp_c`, `nacelle_position_deg`, `pitch_a_deg`, `power_kw`, `wind_direction_deg`, `wind_speed_ms`.

2. Turbine scorecard

Turbine	Flag	Confidence	Non-production at wind (inferred)	Abnormal oil days, final yr	Under-perf. days, final yr	Pump pressure: base to last (bar)	Oil-temp model error
LHB01	GREEN	Medium	0.8%	-	0%	-	-
LHB02	GREEN	Medium	0.6%	-	0%	-	-
LHB03	GREEN	Medium	0.5%	-	0%	-	-
LHB04	GREEN	Medium	1.2%	-	0%	-	-

Ratings, reasons and notes

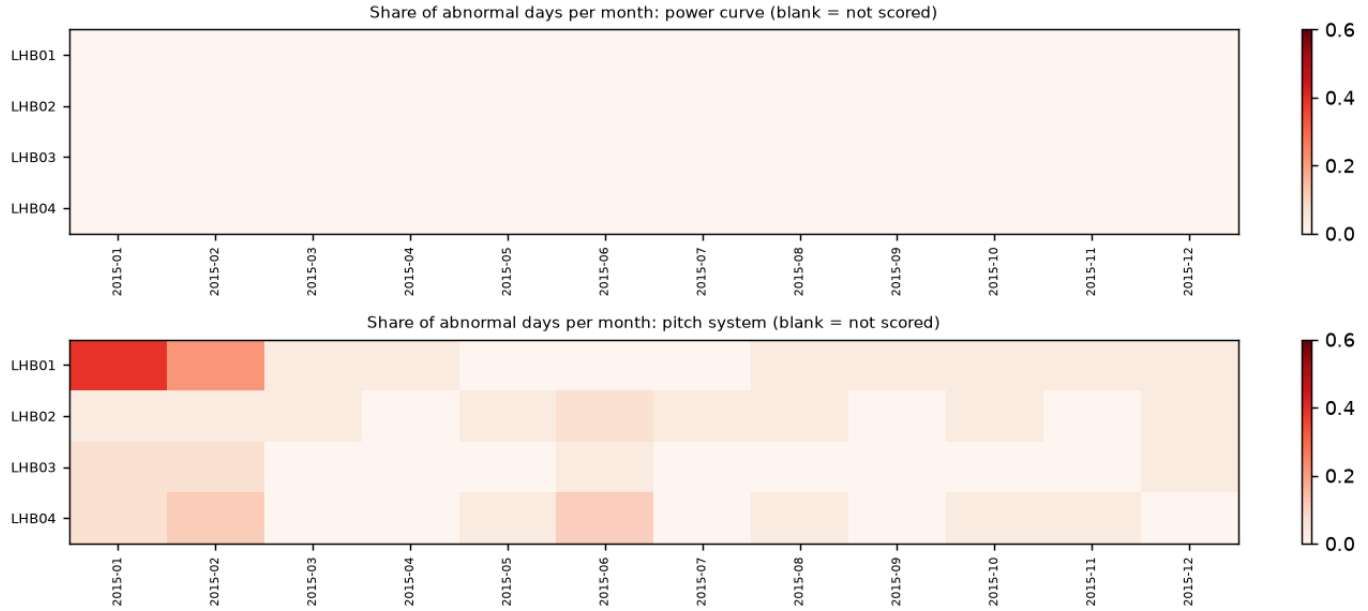
- **LHB01** GREEN (confidence Medium: 24 months of history, models trained on earlier years, no alarm log (trip rules unavailable)). No rating trigger.
- **LHB02** GREEN (confidence Medium: 24 months of history, models trained on earlier years, no alarm log (trip rules unavailable)). No rating trigger.
- **LHB03** GREEN (confidence Medium: 24 months of history, models trained on earlier years, no alarm log (trip rules unavailable)). No rating trigger.
- **LHB04** GREEN (confidence Medium: 24 months of history, models trained on earlier years, no alarm log (trip rules unavailable)). No rating trigger.

3. Model-based health (normal-behaviour residuals)

Each turbine gets its own models that predict a signal (oil temperature, bearing temperature, pitch angle, power...) from operating conditions. A day is *abnormal* when the turbine's median deviation from its own model is beyond 3 robust standard deviations in the harmful direction. *Chronic* means at least 30% abnormal days in a month.

Turbine	Training scheme	Models	Window scored	pitch system: abnormal days final yr / chronic months	power curve: abnormal days final yr / chronic months
LHB01	walk-forward	2	50%	7% / 1	0% / 0
LHB02	walk-forward	2	49%	3% / 0	0% / 0
LHB03	walk-forward	2	50%	2% / 0	0% / 0
LHB04	walk-forward	2	50%	3% / 0	0% / 0

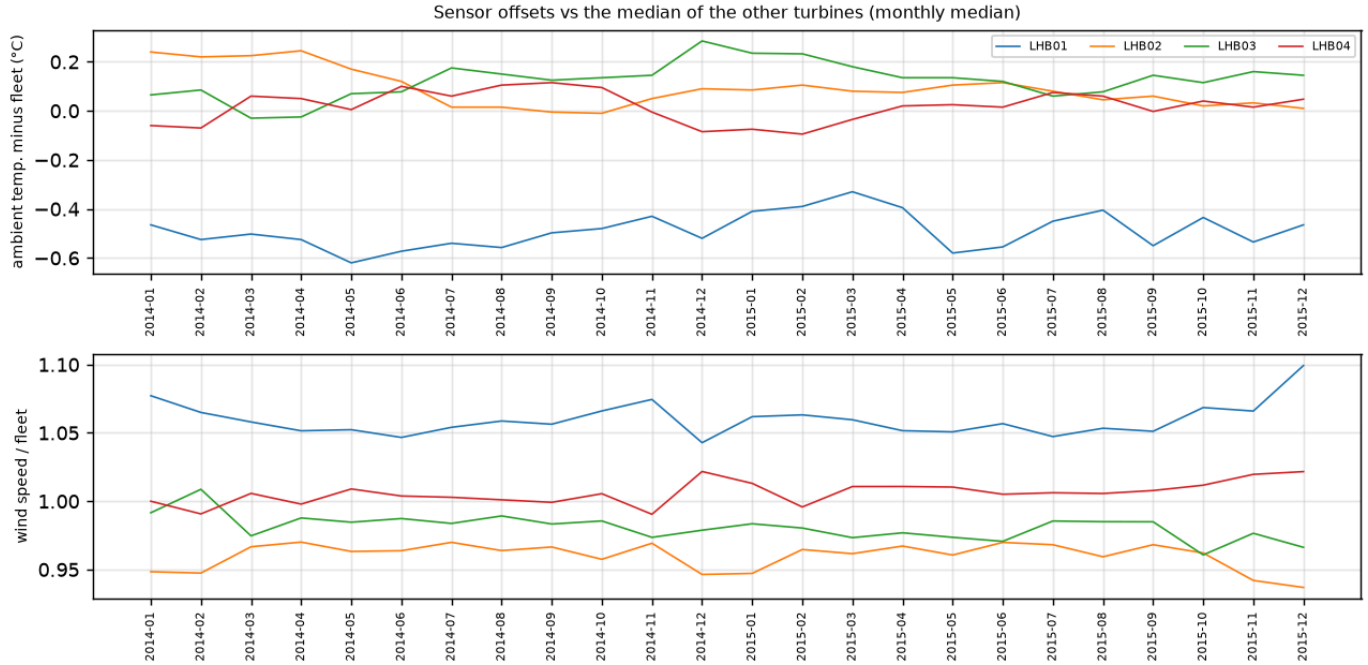
No turbine has 3 or more consecutive chronic months in the window.



5. Sensor drift and stuck sensors

Drift = change of a turbine's offset to the median of the other turbines, last 3 months vs before the window (thresholds: 1.5 °C, 5% wind speed).

Turbine	drift ambient_temp_c	drift wind_speed_ms	Stuck sensors
LHB01	+0.06 °C	1.3%	-
LHB02	-0.20 °C	-2.1%	-
LHB03	+0.08 °C	-2.1%	-
LHB04	+0.01 °C	1.8%	-



6. Stop and fault history

The export has no alarm or status log. Stops were inferred from SCADA: power below 1% of rated while the wind is between 4.5 and 25 m/s. That catches downtime, but cannot tell a fault from a grid outage, maintenance or curtailment, so no stop/fault history, availability or trip analysis is possible.

7. Methods, thresholds and limits

- **Pipeline.** `windml report` reads the raw export only through the mapping (`windml.ingest`), cleans and labels each turbine (QC flags, short-gap interpolation, alarm-log stop labels), trains the models, computes the metrics and writes this report. Every number here is reproducible from the mapping and the raw files.
- **Window.** The last 24 months of each turbine; *final year* = its last 12 months.
- **Models.** LightGBM per turbine and signal on healthy operating rows (no stop within 6 h, no fault, not in the 30 days before a major fault, QC clean). With 2+ calendar years, each year is scored by a model trained only on earlier years (3-day embargo, training must cover all 12 months). With 12-23 months, the record is cut into 6 blocks scored by models trained on the other blocks (cross-fit). Under 12 months, no model.
- **Ratings.** Computed at each month-end from data up to that date (latest month-end shown here). AMBER when: the full-load gear-oil pump pressure (median of the last 15 days) is >= 1 bar above the turbine's own baseline (days 60-365 before) AND >= 5 robust standard deviations above the other turbines' rise that month; >= 3 oil-system trip episodes in 6 months; >= 6 pitch trip episodes in 6 months; an ambient-temperature or wind-speed sensor drifting vs the fleet over the last 3 months (1.5 °C, 5%); or a monitored sensor frozen on more than 2% of the last 90 days. RED when the pump rise is >= 2 bar; or a trip rule has held 2 month-ends in a row with >= 3x the amber count and a trip in the last 30 days; or two independent systems are amber at once. Shown as notes but never colour-setting, because they had no reliable lead on events in the validation: low availability (it describes the past), model-based abnormal-day rates (oil, pitch, main bearing, power curve) and missing data. A trip episode merges trips less than 24 h apart.
- **Validation.** Retrospective check on Kelmarsh and Penmanshiel (21 turbines). Thresholds selected on 2017-2020, checked on 2021-2024 (792 turbine-months): pump build-up → oil trip or service stop: 58% of 12 flagged turbine-months vs 55% by chance; repeat oil trips → further oil trips: 63% of 75 flagged turbine-months vs 32% by chance; repeat pitch trips → further pitch trips: 67% of 6 flagged turbine-months vs 36% by chance. Any amber or red: 76% followed by a logged event within 90 days vs 50% by chance (recall 19%); red alone 65% vs 58% on only 17 turbine-months, so red marks strength and persistence more than a higher probability. Held-out ratings: 89% green, 9% amber, 2% red. Consecutive months overlap, so the effective sample is much smaller than the row counts; details in `reports/fleet_proof/calibration.md`.
- **Confidence.** Points for history (3+ years: 2, 2-3 years: 1), coverage >= 95%, models trained on earlier years, an alarm log, and a fleet of 4+ turbines (peer comparison). High >= 5, Medium 3-4, Low <= 2.
- **Limits.** The alarm-log classes come from the mapping's rules; trip detection uses Senvion-style oil/pitch codes and messages (other vendors need `class_rules`). Full-load oil medians assume one oil circuit design across the fleet. Models learn whatever was normal in their training data, including a fault that was already present. Sensor drift is relative to the other turbines, so a fleet-wide drift is invisible. The rating thresholds were calibrated on two UK farms of one turbine family (Senvion MM92/MM82); on other fleets the peer-relative logic still applies but the hit rates are unknown. Ratings look back up to 6 months and only at what SCADA and the alarm log show: inspections, oil analyses, vibration (CMS) data and work orders are not used, so a recent repair or a fault the sensors cannot see changes nothing here.
- **Data.** Licence Ouverte / Open Licence 2.0 (Etalab; compatible with CC BY 4.0), ENGIE Group.